

Design of Experiments (DoE) in Pharmaceutical Technical Development

Design of Experiments (DoE) has become an indispensable, systematic framework in pharmaceutical technical development for efficiently navigating complex, multivariable systems. In process development, DoE is utilized to identify critical process parameters (CPPs) to optimize yield and product quality, while simultaneously establishing a robust design space resilient to inherent manufacturing variability. Furthermore, applying DoE to analytical development ensures the optimization of method accuracy and precision.